

Puzzles to Puzzle You (Remastered)

Welcome to the second iteration of *Puzzles to Puzzle You (Remastered)* by Rani Rachavelpula, in which I will discuss the previous puzzle (Dogs in Pursuit) and pose the next one.

I want to begin by noting how delightful it was to think about four dogs running after one another and to know that my brilliant friends were thinking the same. This alone proves to me that abstract objects are not casually inert. All to say, I learned a lot from this very first puzzle and I hope you did too. Thank you for being a part of my revisionist puzzles community ♡

Puzzle #1 (Dogs in Pursuit)

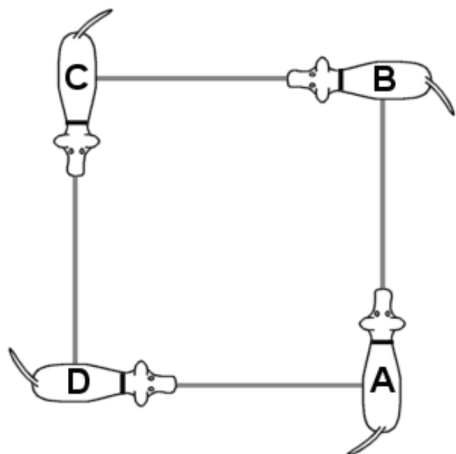
Dogs in Pursuit Four dogs are located at the corners of a square. On command they start moving with constant speed v , each towards its counterclockwise neighbor. If each side of the square has length l ,

1. How long will it take for the dogs to meet?

$$\boxed{\frac{l}{v}}$$

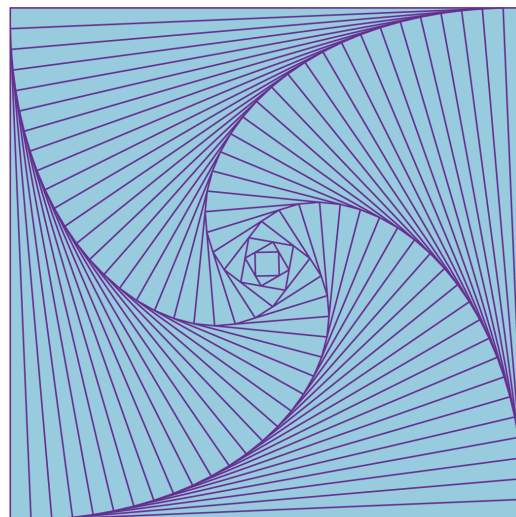
2. What distance will each have covered?

l = the length of the square



Solution (Dogs in Pursuit)

What a wonderful result! As dog B moves, dog A begins to turn and run in a spiral, where the arclength of that spiral is equal to the side length l — their original distance apart!



Dogs' pursuit curves.

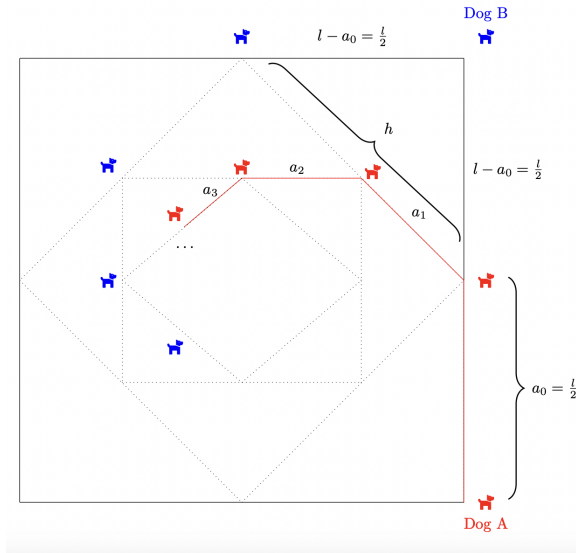
This puzzle belongs to a class of so-called *pursuit problems*. Each dog's path constitutes a *pursuit curve*. There are many other well-known pursuit problems, including the dog and goose circular pursuit problem.

Proof

My solution begins with considering the series approximation of the arclength of the curve. The symmetry of the problem allows us to focus our attention on a single pair of dogs. Let's choose dogs A and B. Let's first calculate the distance dog A will travel if it is not allowed to turn continuously, but only once it gets halfway.

(n=2) Dog A begins by running halfway up the original square, turns, runs halfway, turns, runs halfway, turns, and so on... The total distance dog A runs is just the sum of these half-distances. Let a_0 be the first half-distance, a_1 the next, and so on such that

$$\text{Total Distance} = a_0 + a_1 + a_2 + \dots$$



Dog A's path as an infinite series $\sum a_i$.

Notice each half-distance depends only on the one immediately before it. If a_i is the i^{th} half-distance, then by the Pythagorean Theorem, we have $a_{i+1} = \frac{\sqrt{2}}{2} \cdot a_i$. Thus, we can model the dog's path as a geometric series where

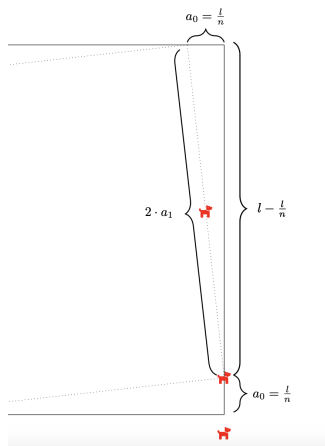
$$\text{Total Distance} = \sum_{n=0}^{\infty} a_0 r^n$$

In this case, $a_0 = \frac{l}{2}$ and $r = \frac{\sqrt{2}}{2} < 1$. Hence, the series converges with sum

$$\frac{a_0}{1-r} = \frac{l}{1-\frac{\sqrt{2}}{2}}$$

Now, we need to generalize this to the case where the dog is allowed to turn continuously.

(n arbitrary) Now suppose the dog turns after running an n^{th} of the distance. Then, the first term in our series is $a_0 = \frac{l}{n}$. The second term can be found using the Pythagorean Theorem as before.



Thus, we have

$$\begin{aligned} a_0 &= \frac{l}{n} \\ a_1 &= \frac{h}{n}, \text{ where } h = \sqrt{a_0^2 + (l - a_0)^2} \\ r &= \frac{a_1}{a_0} = \frac{h}{l} = \sqrt{\left(\frac{1}{n}\right)^2 + \left(1 - \frac{1}{n}\right)^2} \end{aligned}$$

Now, all we have to do is calculate the sum of the series as $n \rightarrow \infty$. We have

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_1}{1-r} &= \lim_{n \rightarrow \infty} \frac{l}{n(1 - \sqrt{(\frac{1}{n})^2 + (1 - \frac{1}{n})^2})} \\ &= \lim_{n \rightarrow \infty} \frac{l}{n(1 - \frac{\sqrt{(n-1)^2 + 1}}{n})} \\ &= l \cdot \lim_{n \rightarrow \infty} \frac{1}{n - \sqrt{(n-1)^2 + 1}} \\ &= l \cdot \lim_{n \rightarrow \infty} \frac{n + \sqrt{(n-1)^2 + 1}}{n^2 - ((n-1)^2 + 1)} \\ &= l \cdot \lim_{n \rightarrow \infty} \frac{n}{2n-2} + \frac{\sqrt{n^2 - 2n + 2}}{2n-2} \\ &= l \cdot \lim_{n \rightarrow \infty} \frac{n}{2n-2} + \sqrt{\frac{n^2 - 2n + 2}{4n^2 - 2n + 4}} \end{aligned}$$

As $n \rightarrow \infty$, both the left and right summand go to $\frac{1}{2}$. Thus,

$$\begin{aligned} \text{Total Distance} &= l \cdot \left(\frac{1}{2} + \frac{1}{2}\right) \\ &= l \end{aligned}$$

$$\begin{aligned} \text{Total time} &= \frac{\text{total distance}}{\text{speed}} \\ &= \frac{l}{v} \quad \square \end{aligned}$$

Armchair ‘Proof’ Given the elegant solution, I began thinking whether there was an intuitive way of seeing why the total distance ends up being the length of the square l .

I came up with the following valid argument though I am not sure if it is sound: Let d be the total distance ran by dog A. Since $l, d \in \mathbb{R}$, exactly one of the following must be true: (1) $d < l$, (2), $d > l$, or (3) $d = l$.

Since the dogs all run at constant speed, the first case is only possible if dog B is running closer to dog A. Similarly, the second case

is only possible if dog B is running further away from dog A. Since the dogs all go at the same constant speed, these two cases are only possible if and only if at some time t , dog A turns by an angle of θ_1 and dog B turns by an angle of θ_2 where $\theta_1 \neq \theta_2$. But this is impossible given the symmetry of the problem. The dogs will always run orthogonal to one another. Since metrics supervene on inner products, it follows that $\theta_1 = \theta_2$, so the first two cases are impossible and $d = l$, as desired.

Shout Out (Noah) First shout out goes to my friend Noah who did what I could never do: Build a simulation. In particular, Noah created a computer program which simulates the dogs' trajectories over time, subject to the relevant constraints. I admit, upon hearing that Noah took this approach, I was disappointed. I considered it cheating, as it seemed to me—a girl who cannot code—to disrespect the ethos of the newsletter. However, upon hearing Noah describe his code, I realized his approach was basically the same as mine. In fact, he was much more clever by letting someone else (in this case, something else) do the perfunctory number-crunching, focusing his own attention on the fundamental structure of the problem. This is an important reminder to always keep an open mind and to avoid assuming the worst in others, especially your friends.

Shout Out (Andrew) My dear friend Andrew took a differential equations route, giving a solution involving the distance formula. With this approach, Andrew calculated the problem for not just four dogs in a square, but for n dogs in an n -gon of side length l . In the general case, we have

$$\text{Total Distance} = \frac{l}{1 - \cos\left(\frac{2\pi}{n}\right)}$$

From Andrew's solution, we see that the square is the most efficient n -gon. Thanks Andrew!

Puzzle #2: Dido's Problem

The next puzzle dates back to antiquity. In the *Aeneid*, the Roman poet Virgil (70-19 B.C) tells the story of the Phoenician Queen Dido. After her brother Pygmalion killed her husband (who was also her uncle), Dido fled to North Africa where she asked the local leader Yarb for some land. He agreed to give her as much land as could be enclosed by the hide of a bull.

Dido cut the hide into narrow strips, tied the strips together, and began contemplating which shape would maximize the area bounded by the strip. Utilizing the seashore as part of the boundary, Dido successfully laid her strip to enclose the maximum possible area. Within the enclosed area was built the city of Carthage.



Queen Dido founds Carthage

We can pose Dido's problem as follows:

1. Determine from all simple closed planar curves of the same perimeter p the one that encloses maximal area.
2. Prove its maximality (bonus).

Though Dido intuited the correct answer, a proof evaded mathematicians until the 18th century. Today, Dido's problem continues to be relevant to contemporary research in calculus of variations, PDEs, and differential geometry. ¹ Go Dido!

¹In 1937, mathematician Karl Menger wrote: "The first human being to solve a problem of calculus of variations seems to have been Queen Dido of Carthage."